

Contextual Coherence in Sherlock Holmes Paragraph Continuation via GPT-2 Fine-Tuning

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Abstract

We study next-paragraph generation in Sherlock Holmes stories with a controlled set of GPT-2 fine-tuning experiments. Given the first k paragraphs of a passage, the model predicts the next one. We keep the backbone fixed at `distilgpt2` and vary four design choices: prompt format, context length, tuning strategy, and an auxiliary ranking loss. We evaluate five main experiments (E1–E5) with target-conditioned perplexity, ROUGE-L, BERTScore, and a simple entity-overlap diagnostic under three decoding seeds. Structured formatting helps a little. Moving from two paragraphs to four helps more, and E3 gives the best overall result. LoRA falls well behind full fine-tuning in this setting. The ranking objective does not beat the strongest baseline. Appendix A reports a retrieval comparison, but it does not beat E3.

1 Introduction

Neural generators still struggle once coherence has to stretch past a sentence or two. A model can write fluent text and still lose the scene, shift the speaker, or contradict what just happened. Paragraph continuation makes that problem hard to ignore. The system has to write the next paragraph from a short but information-rich discourse history, so fluency alone is not enough.

The Sherlock Holmes stories give us a good testbed for this setting. The books share a stable narrative world, a small set of recurring characters, and a fairly consistent register. That makes the failures easy to see. If a continuation drifts, the reader notices.

Our question is narrow on purpose: which modeling choices matter most for paragraph continuation in this setting? We leave retrieval and prompt tricks out of the main story. Instead, we compare the choices that sit closest to the model itself: how we serialize context, how much context we give it, whether LoRA can match full fine-tuning, and whether an auxiliary ranking loss helps.

The mainline follows four pairwise comparisons. E1 versus E2 asks whether structured paragraph markers help more than plain concatenation. E2

versus E3 asks whether a longer context window helps. E3 versus E4 compares full fine-tuning with LoRA under the same long-context structured setup. E3 versus E5 tests the auxiliary ranking objective against the strongest no-auxiliary baseline. We also run E5W as a robustness check for the auxiliary weight, but we do not treat it as a separate mainline result.

The results line up cleanly. Structured input helps a little. Longer context helps more, and E3 is the strongest model in the mainline. LoRA performs worse than full fine-tuning here. The auxiliary ranking loss does not improve on the strongest baseline. Retrieval stays in the appendix because it does not change that picture.

2 Related Work

Our work sits closest to the literature on autoregressive transformers for conditional text generation. These models fit continuation tasks naturally because they generate left to right and adapt well through fine-tuning [7, 5]. GPT-style models matter here for a practical reason too. They are strong enough to show meaningful differences between design choices, but small enough to run in a controlled Colab workflow.

The paper also connects to long-form and story generation, where coherence remains the central difficulty. Prior work shows the same pattern again and again: models can look fluent locally and still lose the thread over longer spans [1, 2]. Paragraph continuation gives us a smaller and cleaner version of that problem.

Parameter-efficient adaptation is the third obvious reference point. Methods such as LoRA try to preserve most of the quality of full fine-tuning while updating far fewer parameters [3]. That trade-off is worth testing in our setting because compute is limited and the backbone is small.

Our last connection is to work that goes beyond plain next-token loss. If token prediction alone leaves coherence on the table, an auxiliary objective might help. That is the motivation behind E5. We test a simple ranking signal rather than a more elaborate coherence model. Retrieval belongs in the

background here, but only as appendix material.

3 Methodology and Architecture

3.1 Task Formulation

We treat the task as conditional next-paragraph generation. Each example contains the previous k paragraphs and the next paragraph from the same source text. The loss is token-level, but the target behavior is not. We want a continuation that fits the local scene, keeps the narrative state intact, and does not wander off stylistically or semantically.

3.2 Backbone Model

All mainline experiments use `distilgpt2` as the shared backbone. We chose it for control as much as for convenience. The model is small enough to train repeatedly in Colab, but still strong enough to make the comparisons matter. Keeping the backbone fixed across E1–E5 lets us attribute most differences to the design choice under test rather than to a change in scale.

3.3 Input Representation

Our first design choice is prompt format. In the plain setting, we concatenate context paragraphs directly. In the structured setting, we insert explicit paragraph markers. The idea is simple: if the model sees discourse boundaries more clearly, it may keep better track of paragraph roles and local state. We test that in E1 versus E2.

3.4 Context Window

Our second choice is context length. The short setting uses $k = 2$ paragraphs and the long setting uses $k = 4$. More context may help the model track the scene, characters, and local plot state, but it also pushes against the token budget. We isolate that trade-off through E2 versus E3.

3.5 Fine-Tuning Strategy

The third comparison is full fine-tuning against LoRA. Full fine-tuning updates the whole model. LoRA trains a small set of low-rank adapters while leaving the base model mostly fixed. We use E3 versus E4 to see whether that efficiency trade-off still looks attractive in a small-model literary task.

3.6 Auxiliary Objective

The last comparison changes the objective. The baseline uses standard causal language-model loss

Table 1: Dataset split summary. Fill the exact train and validation counts before submission.

Split	Samples
Train	5285
Validation	220
Test	663

Table 2: Main experiment matrix.

ID	Goal	Setup
E1	Baseline	$k = 2$, plain, full, no aux
E2	Structure	$k = 2$, structured, full, no aux
E3	Context length	$k = 4$, structured, full, no aux
E4	Fine-tuning	$k = 4$, structured, LoRA, no aux
E5	Objective	$k = 4$, structured, full, ranking

on the target continuation. The auxiliary setting adds a ranking loss that should push the model to score the true next paragraph above sampled negatives. We test that idea through E3 versus E5.

4 Training Methods and Experimental Setup

4.1 Dataset and Preprocessing

We use four public-domain Sherlock Holmes texts from Project Gutenberg: *The Adventures of Sherlock Holmes*, *The Memoirs of Sherlock Holmes*, *The Hound of the Baskervilles*, and *The Return of Sherlock Holmes*. The preprocessing pipeline removes boilerplate material, segments each book into paragraphs, filters out unsuitable paragraphs, and converts the cleaned corpus into supervised (*context*, *target*) continuation examples. Each example consists of the previous k paragraphs and the next paragraph from the same book.

The processed dataset contains 5285 training examples, 220 validation examples, and 663 test examples. All reported numerical results use this fixed split.

4.2 Main Experiment Matrix

The main experiment line is designed to isolate one design factor at a time, as shown in Table 2.

An additional run, E5W, repeats E5 with a larger auxiliary weight. This run is not treated as part of the core mainline narrative. Instead, it functions as a robustness check for the auxiliary setting and supports the final E5 selection using validation-side model comparison.

4.3 Training Setup

Across the mainline experiments, we keep the core hyperparameters fixed: model = `distilgpt2`, epochs

= 3, batch size = 2, gradient accumulation steps = 4, learning rate = 5×10^{-5} , warmup steps = 20, maximum sequence length = 512, and training seed = 42. Holding these settings constant helps keep the pairwise comparisons interpretable. Model selection is based on validation performance rather than test-set outcomes. For the E5 versus E5W comparison, we use validation main loss rather than test metrics to decide which auxiliary-weight variant should represent the mainline auxiliary configuration.

4.4 Evaluation Protocol

We report automatic metrics only in the current report version. The headline metrics are target-conditioned perplexity, ROUGE-L [4], and BERTScore F1 [8]. We additionally report entity overlap as a diagnostic metric. It is useful for rough consistency inspection, but it is not treated as the sole basis for major claims because it is a lightweight heuristic rather than a full semantic evaluation.

Generation-based metrics are evaluated with three fixed random seeds (13, 42, and 2026). We report the mean for all metrics and the standard deviation for generation-based metrics. Perplexity is deterministic for a fixed checkpoint and is therefore reported as a mean or single-value quantity across seeds. This evaluation protocol is intended to distinguish stable model differences from small amounts of sampling variance.

4.5 Scope Note

Human evaluation remains deferred. We did run retrieval, but we report it only in Appendix A. The main claims of the paper rest on E1–E5.

5 Numerical Results

5.1 Main Results

Table 3 summarizes the mainline results. Test metrics are reported on the full held-out test set of 663 samples.

For robustness checking, E5W achieves a validation main loss of 3.1062, validation perplexity of 25.8169, test perplexity of 28.5020, ROUGE-L of 0.1111 ± 0.0007 , BERTScore F1 of 0.8343 ± 0.0003 , and entity overlap of 0.1231 ± 0.0041 . Because it does not improve the validation-side selection criterion relative to E5, it is not used as the main auxiliary result in Table 3.

5.2 Pairwise Analysis

E1 vs. E2: input structure. E2 is a little better than E1, but the gap is small. ROUGE-L rises from 0.1113 to 0.1124, BERTScore moves from 0.8341 to

0.8343, and validation main loss drops from 3.1019 to 3.0998. That is enough to say the structured format helps. It is not enough to claim a large change in quality.

E2 vs. E3: context length. E3 is the best model in the mainline. It has the lowest validation main loss (3.0860), the lowest validation perplexity (25.0467), and the lowest test perplexity (27.6525). ROUGE-L and BERTScore stay close to E2, so the gain shows up more clearly in the likelihood-based metrics than in surface-overlap metrics. Even so, the direction is consistent enough to support the longer context setting.

E3 vs. E4: full fine-tuning vs. LoRA. This is the cleanest result in the paper. Under the same long structured setup, LoRA falls well behind full fine-tuning. Validation main loss rises from 3.0860 to 3.2966, test perplexity jumps from 27.6525 to 32.6382, and the overlap metrics drop as well. In this task, with this backbone, full tuning is simply the better choice.

E3 vs. E5: auxiliary ranking objective. E5 does not beat E3. Its validation main loss is higher (3.1022 vs. 3.0860), its validation perplexity is higher (25.6313 vs. 25.0467), and its test perplexity is higher (28.2869 vs. 27.6525). ROUGE-L and BERTScore barely move, while entity overlap drops. The safest reading is that the ranking loss does not help in its current form.

5.3 E5 Robustness Note

We also compare E5 with E5W, which uses a larger auxiliary weight. E5W is slightly worse on validation main loss (3.1062 vs. 3.1022). The difference is small, so E5W still works as a robustness check. It does not change the conclusion.

5.4 Qualitative Examples

The qualitative examples should do one job: make the numerical differences easier to see. The most useful set would include one case where E3 keeps the scene or character state better than E4, one case where E3 and E5 read similarly fluent but E5 offers no real coherence gain, and, if space allows, one case where the structured prompt helps more than the plain one.

5.5 Limitations

This study has clear limits. We rely on automatic evaluation, so we do not yet have a human judgment of narrative quality. We also work in one literary domain with a small backbone, which limits how far

Table 3: Mainline results on the held-out test set. Perplexity is deterministic for a fixed checkpoint, while ROUGE-L, BERTScore F1, and entity overlap are reported as mean $\pm$ standard deviation over three decoding seeds.

Experiment	Validation Main Loss	Validation PPL	Test PPL	ROUGE-L	BERTScore F1	Entity Overlap
E1	3.1019	25.1105	27.8093	0.1113 ± 0.0009	0.8341 ± 0.0004	0.1435 ± 0.0031
E2	3.0998	25.1061	27.8216	0.1124 ± 0.0002	0.8343 ± 0.0001	0.1434 ± 0.0025
E3	3.0860	25.0467	27.6525	0.1115 ± 0.0003	0.8343 ± 0.0004	0.1266 ± 0.0008
E4	3.2966	30.3848	32.6382	0.1097 ± 0.0005	0.8329 ± 0.0004	0.1275 ± 0.0039
E5	3.1022	25.6313	28.2869	0.1113 ± 0.0012	0.8343 ± 0.0002	0.1207 ± 0.0014

we should generalize. One more caveat matters for the core result: the strongest long-context setting still truncates a non-trivial share of examples under the 512-token budget. The token-budget inspection gives truncation rates of about 24.7% on the training split and 17.3% on the evaluation split for the $k = 4$ structured setting. That does not overturn the result that E3 performs best. It does mean the gain from longer context is harder to interpret cleanly than the headline result suggests.

6 Conclusion

We ran a controlled set of paragraph-continuation experiments with a fixed `distilgpt2` backbone and a narrow mainline question. The answer is fairly plain. Structured formatting helps a little. Longer context helps more, and E3 is the strongest system in the mainline. LoRA is clearly worse than full fine-tuning here. The auxiliary ranking loss does not improve on the strongest baseline.

For this task and model scale, the best gains come from giving the model a better view of the preceding text. We do not see the same payoff from switching to parameter-efficient adaptation or from adding a simple auxiliary ranking loss.

Three follow-up directions matter most. A human evaluation would tell us more than automatic metrics can. Larger backbones may change the trade-offs we see here. The appendix retrieval result also points to a narrower lesson: if retrieval is going to help, it will need better design than the simple setup we used here.

7 Statement of Individual Contributions

Replace the placeholders below with the actual team contribution split before submission.

- Member A: dataset construction, preprocessing, and experiment orchestration.
- Member B: model training pipeline and Colab execution.

Table 4: Appendix retrieval comparison against the strongest mainline model.

Model	Val Loss	Val PPL	Test PPL	ROUGE	BERT
E3	3.0860	25.0467	27.6525	0.1115 ± 0.0003	0.8343 ± 0.0004
Retrieval	3.1759	27.4444	29.8028	0.1089	0.8325

- Member C: evaluation pipeline, analysis, and result verification.
- Member D: report writing, literature review, and final integration.

A Retrieval-Augmented Extension

We treat retrieval as an appendix-only comparison, not as part of the main contribution. We ran it as a separate experiment and checked it against the strongest completed mainline model. It did not win. The appendix system uses a lightweight TF-IDF retriever over earlier passages, following a classical sparse retrieval setup [6].

The retrieval result is weaker than E3 on both likelihood-based and overlap-based metrics. We therefore leave it in the appendix and keep the mainline interpretation unchanged.

B Additional Experimental Details

The appendix can later be expanded with the following material:

- exact dataset statistics and split counts;
- token budget and truncation observations;
- generation settings;
- the detailed E5 versus E5W robustness comparison;
- additional qualitative examples.

References

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